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Modelling changes in sleep timing and duration across the lifespan: changes in circadian rhythmicity or sleep homeostasis?

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Summary

Sleep changes across the lifespan, with a delay in sleep timing and a reduction in slow wave sleep seen in adolescence, followed by further reductions in slow wave sleep but a gradual drift to earlier timing during healthy ageing. The mechanisms underlying changes in sleep timing are unclear: are they primarily related to changes in circadian processes, or to a reduction in the neural activity dependent build up of homeostatic sleep pressure during wake, or both?

We review existing studies of age-related changes to sleep and explore how mathematical models can explain observed changes. Model simulations show that typical changes in sleep timing and duration, from adolesence to old age, can be understood in two ways: either as a consequence of a simultaneous reduction in the amplitude of the circadian wake-propensity rhythm and the neural activity dependent build-up of homeostatic sleep pressure during wake; or as a consequence of reduced homeostatic sleep pressure alone. A reduction in the homeostatic pressure also explains greater vulnerability of sleep to disruption and reduced daytime sleep-propensity in healthy ageing.

This review highlights the important role of sleep homeostasis in sleep timing. It shows that the same phenotypic response may have multiple underlying causes, and identifies aspects of sleep to target to correct delayed sleep in adolescents and advanced sleep in later life.

Keywords: models, sleepiness, development, adolescence, circadian, chronobiology

Abbreviations:

DMH: Dorsomedial hypothalamus

EEG: Electroencephalography

KSS: Karolinska sleepiness scale

MA: Monoaminergic

MSF: Mid-sleep on free days

MSLT: Multiple sleep latency test

NREM: Non-rapid eye movement

REM: Rapid eye movement

SCN: Suprachiasmatic nucleus

SWA: Slow wave activity

SWS: Slow wave sleep

VLPO: Ventro-lateral preoptic

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1 Observed changes in sleep across the human lifespan

In many species, the structure, total duration and temporal organisation of sleep exhibit profound changes across the lifespan, and humans are no exception [1, 2].

In early life, sleep is polyphasic, becoming monophasic as children reach primary school age. Timing of sleep for primary school children is often associated with early bedtimes and early wake times. With the onset of puberty the (preferred) timing of sleep is delayed, leading to the characteristic 'owl'-like behaviour of adolescents. Once men and women reach their early 20's this change in timing reverses and there is a drift back to earlier timing. For example, measurements of mid-sleep on free (non-work) days (MSF) indicate that the time of mid-sleep occurs at approximately 3am at age 10, shifts to 5am by the late teens and drifts back to to approximately 3.30am by age 60 [3].

Few studies have quantified age-related changes in sleep duration across the lifespan. In early life, sleep takes up a considerable percentage of the 24-hour day. Studies of self-reported sleep duration/time spent in bed of people from ages 10 to 80, indicate that sleep duration decreases monotonically up until the age of 60 and then levels off, or even increases, in retirement [1, 4, 5]. There are, however, systematic differences between different studies: the American Time Use survey [5] indicates that young adults in the 15-24 age category sleep 10.5 hours on Sundays and that this reduces monotonically to approximately 9 hours a night for the 55-64 age category. In [4], sleep duration on free (non-work) days is seen to decrease from approximately 9.5 hours a night at age 15 to 7.5 hours a night at age 60. A reduction in sleep duration and the maximal capacity for sleep with age is supported by laboratory measurements: when adults were given 16 hours of sleep opportunity per 24 hours for one week, an average sleep duration of 7.5 hours in older adults versus 8.9 hours in young adults was found [6].

Changes in sleep duration and timing are accompanied by changes in sleep structure: in early infancy, sleep is rich in rapid eye movement (REM) sleep with approximately 50% of sleep spent in REM[7]; by the time children reach primary school age REM sleep is reduced to around 20% of sleep time and there is much slow wave sleep (SWS). With the onset of

puberty, SWS and slow-wave activity (SWA) start declining in parallel with synaptic density in cortical layers, with a 65% decrease in SWA observed between the ages of 11 and 17 [8]. This decline of SWS/SWA continues in the middle years of life, with decreases varying in magnitude depending on gender and EEG channel [9, 10]. A continued decline is seen for older adults, but at a slower rate: the SWA decrease seen in a study of adults between the ages of 45 and 90 is reported as about one third of the decline seen between the ages of 20 and 40 [11].

Thus compared to healthy people in their twenties, healthy older people sleep less, go to bed earlier, wake up earlier and their sleep contains less SWS/SWA. These hallmarks of sleep in older adults are accompanied by consistent reporting of reduced sleep consolidation in older individuals with an increase in the number of awakenings, particularly from NREM sleep [12, 13]. This occurs even in healthy people with no sleep complaints or sleep disorders.

The observed age-related changes in sleep, if not driven by societal and environmental pressures, must be the result of underlying physiological changes. Considering age-related change within the context of different functional theories of sleep suggests several possible underlying causes.

1.1 Sleep as a recovery process

One functional theory of sleep conceptualises sleep as a recovery process [14, 15]. In accordance with this view, poor or short sleep at night, both in young and in older people, leads to poor alertness (increased sleepiness), poor attention and impaired cognitive performance during the day [16, 17].

If sleep is primarily a recovery process, then shorter sleep duration at night in older individuals would suggest impaired recovery, perhaps reflecting a reduced ability to activate neural systems generating sleep [18]. Such a reduced ability to activate sleep generating systems with age could explain reductions in SWS, increases in sleep fragmentation (cf [18]), shallower sleep and increases in multiple sleep latency times (MSLT) [20, 21], but would also suggest an increase in daytime sleepiness. On the contrary, when changes in daytime

sleepiness have been assessed using the Karolinska sleepiness scale (KSS), KSS values decrease with increasing age [19]. It is important to note that in older, as in younger people, when sleep fragmentation and reduced SWS are experimentally induced, daytime sleepiness increases [22, 23, 19]. Thus the reduced sleepiness in older people cannot be explained by an inability to respond to disrupted sleep.

An alternative view is that reduction of sleep duration with age reflects reduced sleep 'need', and not a reduced ability to activate sleep. A reduced need for sleep could explain not only the reduction in KSS, but is also consistent with a reduction in SWS and increases in sleep fragmentation and MSLT.

One could argue that reduced KSS is insufficient to distinguish reduced ability to activate sleep-generating systems from reduced sleep need. Indeed, there is no universally accepted marker that can distinguish between these two procesees. However, within the context of aging, not only is KSS reduced but also it has been observed that differences in sleep duration between older and younger participants persist, even when both groups are given a repeated opportunity to sleep for up to 16 hours [6]. Furthermore, there is a reduction in performance decrements during total sleep deprivation with age [24]. Thus it may be argued that reduced physiological need for sleep is the most parsimonious interpretation of the observation of shorter sleep duration in older adults. This interpretation does not require the postulate of a sleep debt, something which cannot readily be measured independently from the ability to sleep.

Overall, the data suggest that, in healthy ageing, sleep need is reduced but that daytime sleep propensity remains responsive to insufficient sleep. In other words, healthy older people, without sleep disorders, do not live under a higher sleep debt and are not sleepier than young people, a stigma often associated with ageing.

Such a reduced sleep need could perhaps be understood within the context of a second functional theory of sleep, where sleep is seen as a use-dependent process that changes with ageing.

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1.2 Sleep as a use-dependent process reflecting changes in plasticity

According to this functional theory of sleep, plastic processes occurring during wake lead to an overall increase in synaptic strength. The role of sleep, and in particular SWS, is then to reduce synaptic strength to a sustainable baseline level and selectively prune the neural network to improve its performance: sleep is the price we pay for plasticity [25, 26].

In support of this functional theory of sleep, wakefulness is observed to accompany high levels of activity in the ascending arousal system, for example, the noradrenergic locus coeruleus [27], and there is a correlation between global noradrenergic activity and SWA: reducing global noradrenergic activity during wakefulness leads to reduced SWA during sleep [28]. Locally intensifying the activity of specific neural networks during wakefulness by, for example, vibrating a hand [29] or learning a motor skill [30], leads to local increases in SWA during subsequent sleep. Whereas locally reducing brain activity, for example by constraining an arm, results in local reductions in SWA [31].

Local cortical synaptic density changes could explain reduced SWS (cf [32, 33, 34, 35, 36, 37]), leading to reduced sleep need. Neuronal changes leading to altered processing of external stimuli could also explain the reduction in sleep consolidation with age. For example, there are observed age-related reductions in sleep spindles (12-14 Hz oscillations) and these have been implicated in sleep consolidation [38, 39, 40]. Within this context, age-related changes in sleep and homeostatic sleep pressure are related to age-related changes in plasticity.

1.3 Sleep as a circadian adaptive rhythm

Finally, a third functional theory of sleep sees sleep as a circadian, adaptive, rhythm of inactivity [41]. This theory sees sleep primarily as a daily rhythm enabling animals to conserve energy during times of cold, high predation risk or low food availability. This rhythm has evolved as a consequence of the Earth's rotation around its axis, leading to daily oscillations in light and temperature. Within this theory, sleep is not seen as having a particular relation to brain function. Circadian rhythmicity is a clock-like process which

modulates sleep propensity, and many other aspects of sleep, such as REM sleep, sleep spindles and to a lesser but significant extent EEG slow waves [42, 43].

At the molecular level, circadian rhythmicity is generated by a group of core clock genes present in nearly all cells. Circadian rhythmicity in mammals is believed to be coordinated by a master circadian pacemaker located in the suprachiasmatic nucleus (SCN) [44, 45]. The SCN receives inputs from the environmental light-dark cycle through a specialised pathway from light sensitive retinal ganglion cells [46], enabling endogenous rhythms to synchronise to 24 hours. Via the dorsomedial hypothalamus (DMH), the SCN inhibits sleep active neurons in the ventro-lateral preoptic area (VLPO), activates orexinergic neurons [47] and limits the secretion of melatonin during the biological night [48].

The net effect is that the SCN modulates the sleep-wake cycle by generating an oscillatory signal that acts in opposition to the homeostatic sleep propensity; promoting wake during the biological day as homeostatic sleep propensity is increasing, and promoting sleep during the night as the homeostatic sleep propensity is decreasing. Sleep remains consolidated throughout the night if the circadian sleep promoting signal is sufficiently strong [49, 45].

Many aspects of circadian rhythmicity are found to correlate with sleep variables. At the gene level, polymorphisms and mutations in clock genes correlate with variation in sleep timing and duration [50, 51, 52] and sleep structure [53, 54, 55]. Individual differences in the timing of molecular rhythms and melatonin concentrations in blood also correlate with individual differences in sleep timing [56]. The inter-individual variation of in-vivo period correlates with individual preferences for timing of sleep in young adults [57, 58, 59], with a shorter endogenous circadian period associated with earlier sleep timing. This correlation is not seen in older people [60].

Age-related changes in circadian regulation have been investigated at many different levels. In humans, the endogenous circadian period does not appear to be markedly affected by age [61], although there are intriguing experiments that show thermolabile serum factors from older individuals can alter *in vitro* periodicity [62]. Age-related changes in patterns of clock gene expression have been reported [63]. Within the SCN, ageing is associated with phase

desynchrony between neurons, resulting in an overall reduced amplitude of multiple unit activity, even though the molecular oscillations within individual neurons may or may not be affected [64, 65, 66, 67]. A reduction in circadian amplitude has been implicated in age-related cognitive decline [68].

Experiments in which the sleep-wake cycle was desynchronized from endogenous circadian rhythms (forced desynchrony) have demonstrated that the basic aspects of circadian regulation of sleep are preserved in old age, but quantitative differences appear. Specifically, in older people: the timing of circadian rhythms, such as the core body temperature and melatonin rhythm, is advanced. Just as in young people, sleep duration is modulated by the circadian phase at which sleep occurs, but in older people total sleep duration and sleep consolidation is reduced at all circadian phases. In fact, older people are much more susceptible to the negative effects of circadian phase misalignment than young people [60, 69, 70, 71].

Hence, seeing sleep as a circadian adaptive rhythm suggests age-related sleep change primarily reflects changes in the circadian process, and functional consequences of these changes for brain function, be it 'recovery' or 'plasticity' may be minimal.

2 Theoretical studies of age-related changes to sleep

Interpretations of age-related changes in sleep have direct consequences for our approaches in dealing with age-related sleep problems. For example, our advice to an older person with short sleep may be very different depending on whether we believe that sleep need declines with age, as opposed to acute sleep need being age-invariant. Typical problems and associated interventions are summarised in Table 1.

In order to provide effective advice, we therefore need a better understanding of the mechanisms by which observed physiological changes translate to changes in sleep timing and duration. Models have been instrumental in providing a framework for understanding sleep-wake regulation, with the seminal two-process model [72, 14, 73, 74, 75, 76] explaining many global aspects of sleep regulation. The two process model provides a quantitative

description of the dependency of sleep duration on circadian phase of sleep onset; the change from monophasic to polyphasic sleep patterns [77]; and the effects of total sleep deprivation on subsequent sleep intensity and duration. Many of its predictions have been confirmed in numerous experiments, and in particular, in forced desynchrony experiments in which the sleep-wake cycle has been desynchronised from endogenous circadian rhythmicity [78, 70, 42].

The two-process model has been combined with data to suggest that the shift of sleep to later times during adolescence could be a consequence of a larger homeostatic time constant, resulting in a slower homeostatic build-up [33]. The two-process model has not been used to model the post-adolescent drift to earlier times *per se*, but, it has been used to model sleep in depression [79]. Like sleep in older adults, a characteristic of sleep timing during depression is earlier wake times. In order to replicate this earlier sleep timing using the two process model, it had to be assumed that sleep onset was fixed by clock time rather than the model. This suggests that the two process model will not be able to explain earlier sleep timing due to age-related change either, and that more sophisticated physiologically-based models are needed.

Recent research on sleep-wake regulation highlights the interaction between sleep active neurons in the VLPO and other areas in the pre-optic area of the hypothalamus, and wake active monoaminergic (MA) brain stem and posterior hypothalamus nuclei [47]. This has resulted in several physiology-based models of the sleep-wake cycle [80, 81, 82] that focus on the principle groups of neurons believed to promote wake and sleep states and link them to homeostatic and circadian processes. In particular, the Phillips-Robinson model and its extensions have been extensively tested against human data and used to describe: the response to impulsive auditory stimuli [83]; measures of sleepiness during sleep deprivation experiments [84, 85]; the effects of caffeine [86]; spontaneous internal desynchrony [87], and the effects of non-rotating and rotating shift schedules [88, 89].

An extended version of the Phillips-Robinson model that includes the interaction of the light-dark cycle with the sleep-wake cycle has been used to investigate changes in diurnal preference [90], where the authors considered ageing within the context of reduced circadian amplitude. However, their results have wider implications as we discuss below, where we use

the extended Phillips-Robinson model to explore the various interpretations of the changes in sleep across the life span.

2.1 The Phillips-Robinson model as a tool to understand age-related sleep change

At the centre of the Phillips-Robinson model are wake active neurons, loosely represented by the MA, and sleep active neurons in the VLPO, which mutually inhibit each other (for a schematic view of the model, see Fig. 1). There is a flip-flop between the firing of wake active neurons and sleep active neurons as the result of a time-dependent sleep drive, D_{VLPO}, consisting of homeostatic and circadian components. The model is entrained to the 24 hour light-dark cycle through the interaction of light with photoreceptors in the eye.

There is a close relationship between the two-process model and the Phillips-Robinson model [77], but with some significant differences. Importantly, the Phillips-Robinson model assigns physiological meaning to some aspects of the two-process model enabling a direct link to be made between physiological changes with changes in the sleep-wake cycle. The version we discuss includes feedback between the light-dark cycle and the sleep-wake cycle; this is a critical factor in understanding how reductions in circadian amplitude impact sleep timing and duration. Specifically, the circadian pacemaker is modelled as a standard oscillator (the van der Pol oscillator) that generates self-sustaining oscillations. This circadian pacemaker has its own natural period of oscillation but is entrained to 24 hours by a forcing term that is derived from the light-dark cycle [91]. Thus the light-dark cycle determines circadian phase relative to clock time, which then helps determine sleep timing. However, sleep timing feeds back on the influence of light on the pacemaker, since it is assumed that the eyelid acts as a gate for light. Consequently, during sleep, no light reaches photoreceptors and the external signal to the circadian pacemaker is turned off; during wake, the amount of light that reaches photoreceptors and the consequent signal to the circadian pacemaker is dependent on clock time. Further details on the model are given in Box 1, where typical time courses of neuronal firing rates, the circadian and homeostatic processes and the light signal to the circadian pacemaker are also shown.

Phillips and Robinson consider the behaviour of their model both as a function of key parameters and the effect of disturbances. Here, we discuss how each of these features can explain changes in sleep-timing and duration across the lifespan.

2.1.1 The effect of changing circadian and homeostatic parameters

Different mechanisms for age-related sleep change can be explored by changing the circadian and homeostatic parameters in the model as in [90] and replicated in Fig. 3.

Changes to two homeostatic parameters have been considered. These are μ, which represents the rate of homeostatic rise during wake, and ν_{vh}, which represents the sensitivity to the homeostatic process (see Fig. 3). Respectively, these represent the build up of sleep need and the ability to activate neuronal systems generating sleep. Reducing either μ or ν_{vh} results: in shorter sleep duration, as seen by both the reduced width of the blue band in Fig. 3(a,c) and the red line in Fig. 3(b,d); a shift in sleep timing, as seen by the change in position of the blue band in Fig. 3(a,c); a smaller difference between upper and lower values of the sleep pressure, as seen by the reduced width of the magenta band, Fig. 3(b,d). The change in sleep timing is most pronounced in the shift to later bedtimes, with a much smaller change in wake times.

The fact that both changing μ and ν_{vh} changes sleep timing is a result of changes to the homeostatic component of the drive, $\nu_{\mathrm{vh}}H$. This term may be reduced either directly by reducing ν_{vh} (meaning sleep pressure is less effective at activating neuronal systems generating sleep and therefore resulting in increased H) or indirectly by reducing μ (a reduction in the rate of homeostatic rise during wake, resulting in a reduced level of H and therefore reduced sleep need). The fact that μ and ν_{vh} have opposite effects on sleep pressure is seen by the position of the magenta band, Fig. 3(b,d), with decreasing ν_{vh} (the sensitivity to the homeostatic process) resulting in higher sleep pressures and decreasing μ (the rate of homeostatic rise during wake) resulting in lower sleep pressures respectively.

The effect of changing two of the circadian parameters, the amplitude, ν_{vc}, and the period T_{c}, has also been considered and is shown in Fig. 3 (e-h). As can be seen in Fig. 3 (e) and

(f), reducing ν_{vc} leads to a large shift in sleep timing, with smaller values of $\nu_{\rm vc}$ resulting in both earlier bedtimes and earlier wake times. There are corresponding small changes in sleep duration and in the average level of the sleep pressure. The most significant effect of the circaidan period $T_{\rm c}$ is also on sleep timing, with smaller periods correlated with earlier sleep timing, see Fig. 3 (g,h). This is in line with observations that clock period is correlated to chronotype [57, 59].

Overall, the dominant effect of changing homeostatic parameters is on sleep duration, with reduced homeostatic drive tending to result in later bedtimes rather than earlier mornings. Whereas, the dominant effect of changing circadian parameters, is to change sleep timing.

In [90], age-related sleep changes are considered to be a result of a reduction in the amplitude of the circadian pacemaker, resulting in earlier sleep timing and reduced average homeostatic sleep pressure as well as a reduced difference between wake time and endogenous circadian phase. However, changing the circadian amplitude cannot explain either the reduction in sleep duration in older adults, or the shift to later sleep times seen in adolescents. In fact, the results in [90] suggest that changing a single parameter cannot reproduce the main characteristics of age-related changes in sleep across the lifespan. As shown in Fig. 3, varying a single parameter monotonically results in a monotonic change to the mid-sleep on free days (MSF), unlike the data of [3].

It therefore seems clear that only changes in both circadian and homeostatic parameters can provide a consistent explanation for changes in sleep timing and duration across the lifespan. Since the evidence from measurements of KSS do not support an increase in homeostatic sleep pressure for older individuals, and there is little evidence for changes in the intrinsic period with age, we focus on the homeostatic parameter μ that describes the rate of homeostatic rise during wake, and the circadian amplitude $\nu_{\rm vc}$.

Comprehensive data on both MSF and sleep duration, from age 10 to 70, are given in [3] and [4] respectively and reproduced in Fig. 4(a). These data show a monotonic decrease in sleep duration and an 'n'-shaped curve for MSF values, peaking at around age 20. The same changes in sleep duration and MSF can be obtained from the extended Phillips-Robinson

model if the two parameters μ and $\nu_{\rm vc}$, representing the rate of homeostatic rise during wake and the circadian amplitude respectively, both reduce with age, as shown in Fig. 4(b). These parameter dependencies are derived as follows.

Running a simulation of the model results in a sleep-wake cycle with a sleep duration and MSF that is dependent on the values of $\nu_{\rm vc}$ and μ. Contour plots for sleep duration and MSF as a function of $\nu_{\rm vc}$ and μ are shown in Fig. 4(c) and (d) respectively. The fact that the contours in (c) are close to vertical shows that varying circadian amplitude has only a small impact on sleep duration (cf Fig. 3(f)). The diagonal nature of the contour lines in Fig. 4(d) shows that MSF monotonically increases with decreasing μ, and decreases with decreasing $\nu_{\rm vc}$ (cf Fig. 3(c) and (e)). For each age, the data for MSF and sleep duration is used to locate a unique point in the $(\mu, \nu_{\rm vc})$-plane for which the model produces the same MSF and sleep duration—the black crosses on Fig. 4(c) and (d) show the positions of these points. This then gives the direct relationship between μ and $\nu_{\rm vc}$ and age shown in Fig. 4(b). Note there is no fitting required here: each pair of values (MSF, sleep duration) in the data maps to a pair of values $(\mu, \nu_{\rm vc})$ which, to within the numerical accuracy of the integration of the model equations, give a sleep-wake cycle with the same (MSF, sleep duration).

This suggests a mechanism for explaining the observed age-related changes in sleep. Namely, that during adolescence there are simultaneous changes in both the circadian amplitude and in the sleep homeostat, which together result in later sleep timing and reduced sleep duration. After adolescence the reduction in circadian amplitude dominates, resulting in progressively earlier timing.

The results in Fig. 4 use the same parameters as in [90] along with the choice relative to clock time that the maximum light intensity occurs at midday. We note that, although the results shown in Fig. 4(d) result in changes in the MSF of the same magnitude as the data, there is a relative shift of two hours between the data and the computed values. Two possible explanations for this are: firstly, that the maximum intensity of daylight is not at 12 noon and is dependent on location within a time zone. Secondly, and more importantly, an individual's exposure to light is far from sinusoidal, and may include limited natural light through day time hours and significant amounts of artificial light, particularly in the evening.

This is likely to offset an individual's light-dark exposure from the natural light-dark cycle.

The results of the simulations shown in Fig. 4 suggest that both changes to circadian and homeostatic variables must occur to explain changes in sleep timing and duration across the lifespan. However, an important aspect of age-related sleep change is reduced sleep consolidation and increased fragmentation in response to stimuli. This can be investigated within the model by delivering perturbations to the system when it is in a sleep state [83]. Note that these perturbations could represent either environmental stimuli, internal stimuli associated with sleep disorders or other problems such as pain, or biological noise.

2.1.2 The effect of disturbances and relation to sleep vulnerability

In the Phillips-Robinson model, the inhibition of the VLPO by the MA, means that it takes a higher value of the VLPO drive to initiate VLPO firing when the system is in a wake state, than to maintain VLPO firing once the system is in a sleep state. This results in hysteresis between wake and sleep states, leading to times of the day when both sleep and wake states are possible. At these times it is sleep/wake history, rather than the value of physiological parameters , that determine whether it is sleep or wake that actually occurs. Within the hysteretic region it is particularly easy to switch between sleep and wake: if asleep, then sleep is vulnerable in the sense that a sufficiently strong stimulus will result in spontaneous waking. The details of this mechanism are further explained in Box 2.

The time spent in the hysteretic region is dependent on the parameters. Consequently, for different parameter values, the same size perturbation applied at the same clock time can have a significantly different effect. This is illustrated in Fig. 5(c) and (d), where the effect of applying a stimulus that momentarily increases the firing rate of the MA at 4.30am every morning is shown for two different values of the parameter μ, corresponding to the 'normal' value ($\mu = 4.4$) and to an 'older' value ($\mu = 3.8$), respectively. In the 'normal' case, the model rapidly falls back asleep following the perturbation. For the reduced value, the perturbation is large enough to shift the model from the sleep to the wake state and no immediate return to sleep is possible. The effect of the early morning perturbation in the long term has a knock-on effect on sleep timing due to the increased light exposure in the

early morning hours, resulting in a slight phase advance for the normal value of μ and a substantial phase advance of sleep-wake timing for the reduced value of μ.

Consequently, the effect of the hysteresis is that sleep in the early morning is particularly vulnerable to disturbances. Furthermore, this region of sleep vulnerability varies, with reducing the strength of the circadian or reducing the homeostatic parameters leading to a larger region of sleep vulnerability (for example, see Fig. 6(a)). Early morning waking results in early morning light exposure and a shift in circadian phase, resulting in earlier bed-times. The net impact on sleep timing is shown in Fig. 6(b), where it can be seen that reducing the homeostatic parameter μ whilst including perturbations can result in an 'n' shaped MSF along with a monotonic decrease in sleep duration, as observed in the data [3, 4]. Consequently, the computed MSF can be mapped to the MSF data to give a relationship between the parameter μ and age, see Fig. 6(d). This suggests that changes in sleep timing and duration can be explained by changing the single parameter related to the rate of homeostatic rise during wake.

We have illustrated model behaviour with a single stimulus of a fixed size, occurring after the circadian minimum, in order to highlight the impact of changes in sleep vulnerability using parameter values as given in [90]. The response of the model to perturbations depends critically on the size of the sleep vulnerability region, a feature that is sensitive to the parameters used to model the neuronal potentials, as well as the distributions in size and timing of any stimulus. Furthermore, a comprehensive study of the effect of perturbations on sleep should include the distinction between REM and NREM sleep. Nevertheless, our analysis illustrates that reducing the rate of homeostatic rise during wake can, not only result in a reduction in sleep duration, but can also effect sleep timing in a way that is consistent with observed changes right from adolescence through to later life.

3 Discussion

It is well established that the structure and timing of sleep change across the lifespan, but a consensus on the underlying mechanisms and the functional significance has not been

reached.

Models can help frame our thinking: particularly where feedback, which can lead to non-intuitive results, is concerned. The Phillips-Robinson model implies that changes in the circadian pacemaker can change sleep timing, but tend not to change sleep duration and means that on its own, changes to the circadian pacemaker may not be able to explain changes in sleep duration across the lifespan. It is then perhaps not surprising that changes in sleep-timing and duration can be explained as the result of a simultaneous reduction in the sleep pressure and in the amplitude of the circadian rhythm, with both changing rapidly during adolescence, perhaps reflecting the significant synaptic re-modelling that occurs during adolescence, and changes in the circadian pacemaker dominating in later years. Importantly, the change in the homeostatic process does not describe a reduced capacity to activate neural systems generating sleep, but instead an age-dependent reduction in the activity dependent build-up of homeostatic sleep pressure. This could be related to age-related changes in plasticity [28]. Such simultaneous reductions could lead to: 1) 'reduced sleep pressure' during wakefulness, an observation consistent with the reduced sleepiness in older people; and 2) reduced SWA during sleep, an observation also consistent with the data.

It is also of interest to consider the role of disturbances within the Phillips-Robinson model. A reduction in the parameter that describes the rate of homeostatic rise during wake results in greater vulnerability of sleep to disturbances. In turn, because of the feedback between the light-dark cycle and the sleep-wake cycle through the gating of the eye, this results in changes to sleep timing. This challenges the common view that changes in sleep timing must be due to changes in the circadian pacemaker. It also suggests that the most parsimonious explanation for age-related changes in sleep is a reduction in acute sleep need. Such an interpretation, of course, does not argue against interventions which may prevent such age-related changes in sleep and associated brain function.

Our review underlines the need to better understand the role of feedback between the different sleep-wake regulatory processes. There is increasing evidence that the SCN is affected by multiple feedbacks: its firing rate is directly modulated by vigilance state, with

reduced firing rate during NREM sleep, compared to wakefulness and REM sleep [92, 93]; the circadian amplitude of rhythms in the SCN are reduced with activity [94]; recent animal and human data show that sleep during the day can reduce circadian amplitude of gene expression in the periphery, including extra SCN brain areas. Light has a strong influence on circadian rhythmicity through both natural and artificial light input to the SCN: artificial light can suppress melatonin, delay sleep onset, and shift the phase of the circadian system [95]. But light availability is coupled to the sleep-wake cycle by the gating effect of the eye. This leads to a complex situation in which changes in the light environment, lead to changes in sleep propensity which in turn lead to changes in light exposure. Whereas early on in adolescence these changes are most likely to impact on the decision to go to bed in the evening, thereby further delaying the clock, later on in life when sleep is shallower, it affects light input in the morning, thereby advancing the clock.

There is also a need to distinguish cause and effect: it may be that altered strength of the homeostat, as reflected in reduced SWS, may result in shorter sleep, leading to altered light exposure that in turn may change the amplitude of many circadian variables [96].

It may also be useful to revisit assumptions in relation to the use of SWS as a marker of the homeostatic process. SWS is deduced from EEG measurements and are an indication of synchronised neuronal activity. Sleep homeostasis has been characterised in dose response studies of the EEG versus wake duration [97], but there are remarkably few studies which have quantified the dose response characteristics in both young and older people [71].

Finally, the further development of physiology-based models in which the interaction and feedback between environmental cycles, neuronal systems generating sleep, and circadian systems are represented in a quantitative manner, has the potential to bring insight to the causes and consequences of changes in sleep timing across the lifespan. It may provide important clues to the aspects of sleep to be targeted to correct extremes in age-related changes to sleep, which can lead to advanced sleep phase disorder later in life and delayed sleep phase syndrome in adolescents. For example, increasing sleep pressure in older people through restriction of time in bed may lead to reduced vulnerability of sleep and consequent later sleep timing. Likewise, strong social early morning zeitgebers in adolescents, combined

with reduced evening light input, may prevent later bedtimes and associated delays in the circadian timing system.

Practice points

Age-related changes in sleep include: reduced sleep duration; reduced slow wave sleep; changes in sleep timing; reduced sleep consolidation. Within the framework of models of sleep, a consistent explanation requires either

- a simultaneous reduction in homeostatic sleep need and in circadian amplitude

or

- a reduction in homeostatic sleep need resulting in reduced sleep consolidation and associated early awakening and light exposure.

Research agenda

Carefully designed interventional studies in young and older people are needed to investigate

- the extent to which changes in sleep duration lead to similar or quantitatively different changes in daytime sleep propensity and waking performance in young and older people,
- the extent to which age-related changes in circadian amplitude originate from within the circadian system itself or whether they are the consequence of reduced feedback from environmental or sleep-wake cycles,
- the role of the environmental light-dark cycle in age-related changes in sleep timing in adolescents and older people,
- the extent to which markers of cortical synaptic strength and neuronal plasticity changes correlate with changes in sleep,
- the degree to which age-related changes in circadian rhythmicity and its amplitude as well as changes in sleep consolidation contribute to cognition.

Models have an important part to play in interpreting data from such studies.

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Table Captions

Table 1. Summary of current interventions for perceived underlying causes of sleep timing problems.

Figure captions

Fig. 1. Schematic presentation of the Phillips-Robinson model. The model consists of a core flip-flop system of mutually inhibitory sleep active (VLPO) and wake active neurons (monaminergic neurons, MA). The firing rate of wake active neurons (Q_{MA}) leads to an increase in homeostatic sleep pressure, where the parameter μ measures the rate of homeostatic rise during wake and is therefore a measure of sleep need. The VLPO is affected by a sleep drive that has homeostatic and circadian components with strengths ν_{vh} and ν_{vc} respectively. Hence, ν_{vh} is a measure of the sensitivity to the homeostatic process, and ν_{vc} is a measure of the circadian amplitude. The circadian pacemaker's endogenous period T_{c} is 24.2 hours but is synchronised to the external light-dark cycle via photoreceptors.

Box 1, Fig. 2. Typical time courses resulting from simulating the Phillips-Robinson model. Neuronal firing rates of the MA and the VLPO are given by Q_{MA} and Q_{VLPO} respectively. During wake, the firing rate of the MA is high and low for the VLPO; whereas during sleep, firing rates are high for VLPO and low for MA (see Fig. 2 (a)). Wake is defined by an MA firing rate above a fixed threshold value. Timing of sleep is indicated by the shaded region.

A flip-flop switching between wake and sleep occurs because a 'drive', D_{VLPO}, see (d), consisting of homeostatic and circadian components, see (b), provides a time dependent input to the VLPO. The homeostatic sleep drive is $\nu_{\mathrm{vh}}H$, where H is the homeostatic sleep pressure and ν_{vh} is a constant which measures the sensitivity to the homeostatic process. The homeostatic sleep pressure is modelled in a similar way to the two-process model, with homeostatic sleep pressure (mostly) increasing during wakefulness and decreasing during sleep (the switch from wake to sleep and from sleep to wake occur within a few minutes of

the maximum and minimum values of the homeostatic sleep pressure respectively). A distinctive feature of the Phillips-Robinson model is that the increase of the homeostatic sleep pressure during wakefulness is directly linked to high MA activity and is governed by two factors: the firing rate in the MA, and the parameter μ that measures the rate of homeostatic rise during wake.

The circadian component of the drive is $\nu_{\mathrm{vc}}C$, where C is the output from the circadian pacemaker. As in [90], the circadian component of the drive is modelled as the approximately sinusoidal output from a van der Pol oscillator representing the endogenous circadian pacemaker with an intrinsic period T_c. The magnitude of the effect of the circadian signal on the VLPO is given by the parameter ν_{vc} and the phase of the circadian signal is determined by the interaction of the sleep-wake cycle with the light-dark cycle. The external light-dark cycle is modelled as a sinusoidal input where, to be specific, we have illustrated model behaviour with a maximal intensity (10000 lux) occurring at a clock time of noon, and minimum intensity (0 lux) at midnight. The external light-dark cycle acts as a forcing term on the endogenous circadian rhythm, through the interaction of light with photoreceptors. The effect of the light is to provide a strong pulse at the change from sleep to wake (c) that both adjusts the phase and entrains the period of the pacemaker to 24 hours. The size of this pulse is dependent on the intensity of light at time of waking.

The parameter δ_{VLPO} represents a background drive to the VLPO and takes the fixed value of 10.2.

In addition to the drive to the VLPO, there is a drive to the MA, D_{MA} motivated by the action of cholinergic and orexinergic nuclei on the MA. When considering age-related changes to sleep, Phillips and Robinson keep D_{MA} fixed, although a time dependent D_{MA} has been used to explain changes in sleepiness in sleep deprivation experiments [84, 85] and to model the role of orexin in narcolepsy [98].

Fig. 3. Sleep timing and sleep pressure as a function of model parameters. Left hand panels show sleep timing as a function of the parameters: sleep times are shaded in blue with the timing of the circadian minimum (- - -) and the timing of mid-sleep (——)

both marked. The right hand panels show the corresponding changes in sleep pressure, H (shaded magenta region) and sleep duration (▬). Panels (a) and (b) show changes for the parameter ν_{vh} that reflects sensitivity to the homeostatic process (ability to activate neuronal systems generating sleep). Panels (c) and (d) show changes for the parameter μ that describes the rate of homeostatic rise during wake (sleep need). Panels (e) and (f) show the effect of changing the circadian amplitude, ν_{vc}. Panels (g) and (h) show the effect of changing the period of the circadian pacemaker, T_{c}.

Fig. 4. Effects on sleep timing and duration of simultaneous changes in the rate of homeostatic rise during wake, μ, and the circadian amplitude ν_{vc}. Data for sleep duration and Mid-Sleep on Free days (MSF) across the lifespan are given in [4] and [3] respectively and are reproduced in panel (a). These show a monotonic decrease in sleep duration and an 'n'-shaped curve for MSF values, peaking at around age 20. The same changes in sleep duration and MSF can be obtained from the extended Phillips-Robinson model if the two parameters μ and ν_{vc}, representing rate of homeostatic rise during wake and the circadian amplitude respectively, both reduce with age, as shown in panel (b). The derivation of the relationship between μ and ν_{vc} and age is explained below with reference to panels (c) and (d).

Running a simulation of the model results in a sleep-wake cycle with a sleep duration and MSF that is dependent on the values of μ and ν_{vc}. Contour plots for sleep duration and MSF as a function of μ and ν_{vc} are shown in panels (c) and (d) respectively.

Each age shown in panel (a) corresponds to a particular sleep duration and MSF. Taking each age in turn, the corresponding values of (μ, ν_{vc}) that result in the same value of sleep duration and MSF in the model can be found—the black crosses on panels (c) and (d) show the positions of these points. This then gives the direct relationship between μ and ν_{vc} and age shown in panel (b).

The red cross on panels (c) and (d) indicate the position of the 'standard' parameters used in the extended Phillips-Robinson model.

Box 2, Fig. 5. The effect of perturbations on the Phillips-Robinson model The mutual inhibition between the VLPO and the MA results in hysteresis between wake and sleep states, leading to a region where both sleep and wake states exist. This can be seen by plotting the firing rate of the MA, Q_{MA}, against the sleep drive, as in Fig. 5 (a). As the drive to the VLPO, D_{VLPO}, oscillates the model switches from wake to sleep and back again, as indicated by the arrows. The region where both wake and sleep states occur for the same drive value is shown in red. Plotting the Phillips-Robinson model in the form of the two-process model illustrates that the hysteretic region corresponds to the region between the upper and lower thresholds, see Fig. 5 (b).

Sleep in the hysteretic region is vulnerable, in the sense that a sufficiently strong stimulus will result in spontaneous waking. Since the time spent in the hysteretic region is dependent on the parameters, for different parameter values the same perturbation can have a significantly different effect. This is illustrated in (c) and (d) where the effect of applying a stimulus that momentarily increases the firing rate of the MA at 4.30am every morning is shown for two different values of the parameter μ corresponding to the 'normal' value ($\mu = 4.4$) and an 'older' value ($\mu = 3.8$), respectively. In the 'normal' case, the model rapidly falls back asleep following the perturbation. For the reduced value, the perturbation is large enough to shift the model from the sleep to the wake state and no immediate return to sleep is possible.

Fig. 6. Vulnerability of sleep and ageing. As described in Box 2, the flip-flop switch results in sleep at the end of the night being 'vulnerable' in the sense that perturbations can result in early awakening. In panel (a), the shaded region shows sleep times as a function of the parameter μ, the rate of homeostatic rise during wake. This is the same as the shaded region in Fig. 3 (a), but now the vulnerable sleep region is distinguished in red. The size of the sleep vulnerability region increases as μ decreases. The average MSF is also marked (——).

The presence of the sleep vulnerability region means that when simulations are performed with random perturbations to the firing rate of the MA, Q_{MA}, the model frequently 'wakes

up' earlier than when no perturbations are applied. This earlier wake time results in earlier light exposure, shifting the phase of the circadian clock, and resulting in earlier bedtimes. The inclusion of random perturbations means that each simulation yields slightly different results. In panel (b), grey crosses indicate the results of 50 simulations for each of a sequence of values of μ: the solid grey line indicates the mean bedtimes and wake times. For comparison, the unperturbed sleep region is shown in blue.

The effect of perturbation is more pronounced for smaller μ because of the increased size of the sleep vulnerability region. Consequently, the shape of the MSF (——) is no longer monotonic.

Since the computed MSF now has an 'n'-shape like the data in [3], the computed MSF data can be mapped to the measured MSF data (black points in panel (c)), resulting in the fit (black line) shown in panel (c). This mapping then gives a relationship between the value of μ and age, as shown in panel (d). This suggests that, if perturbations are included, age-related change in the MSF can be explained by reducing the single parameter μ.

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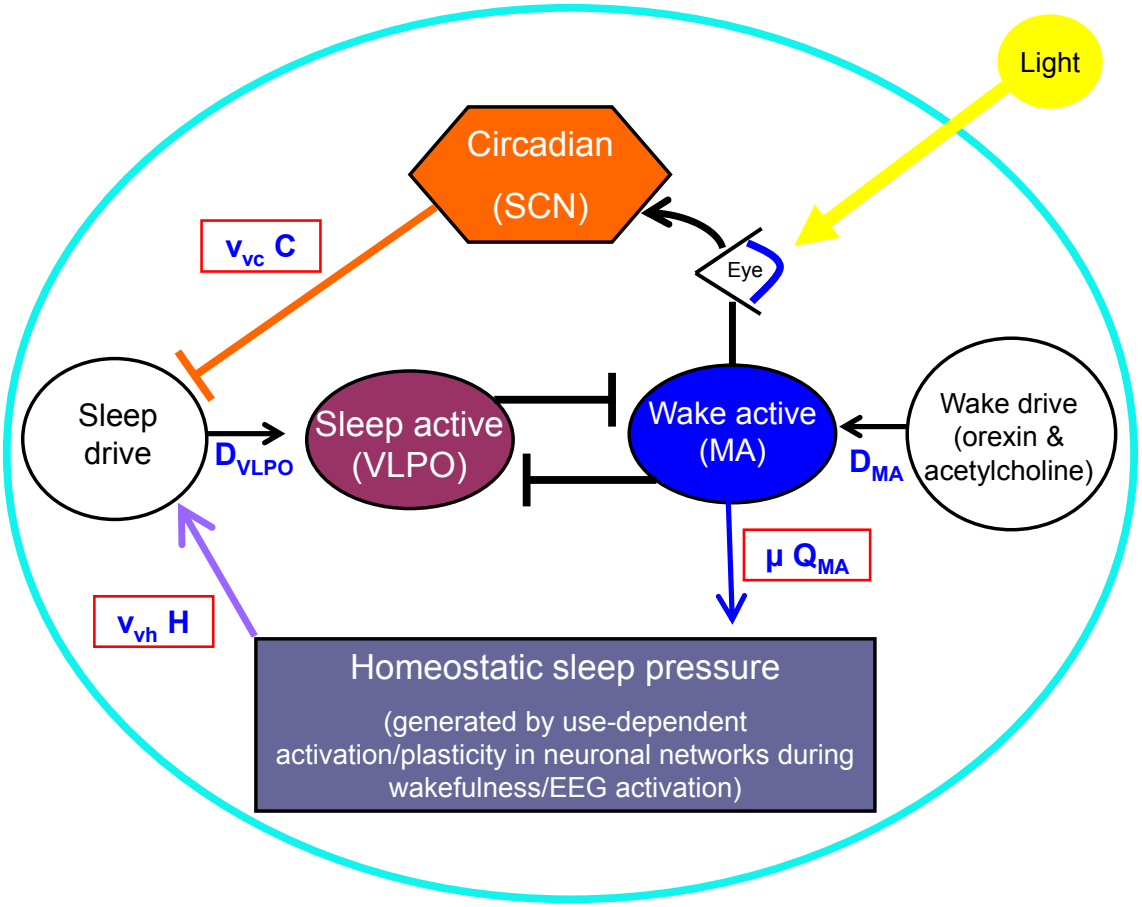
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TABLE 1

Problem	Assumed Primary Cause	Intervention
Adolescents: delayed sleep timing	Lengthening of circadian period	Morning light/delay school
	Reduced homeostatic sleep pressure	Impose early rise time
	Excess social evening light exposure	Reduce social light exposure
Older adults: earlier sleep timing	Shortening of circadian period	Increase light in the evening
	Reduced ability to activate sleep executive mechanisms	Extra time in bed
	Reduced need for sleep	Reduce time in bed
	Increased sleep vulnerability	Reduce time in bed

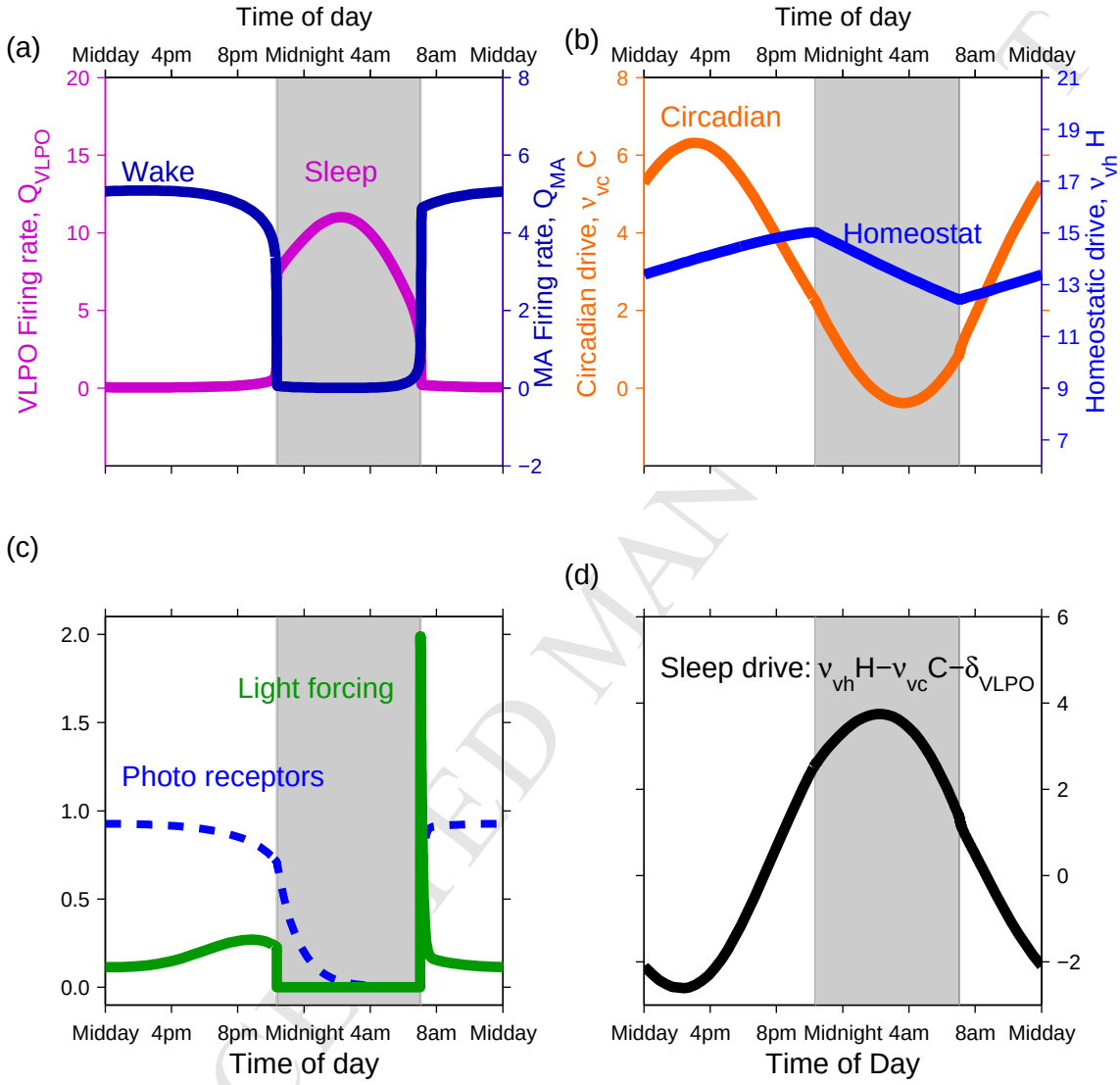
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FIGURE 1



ACCEP

FIGURE 2



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FIGURE 3

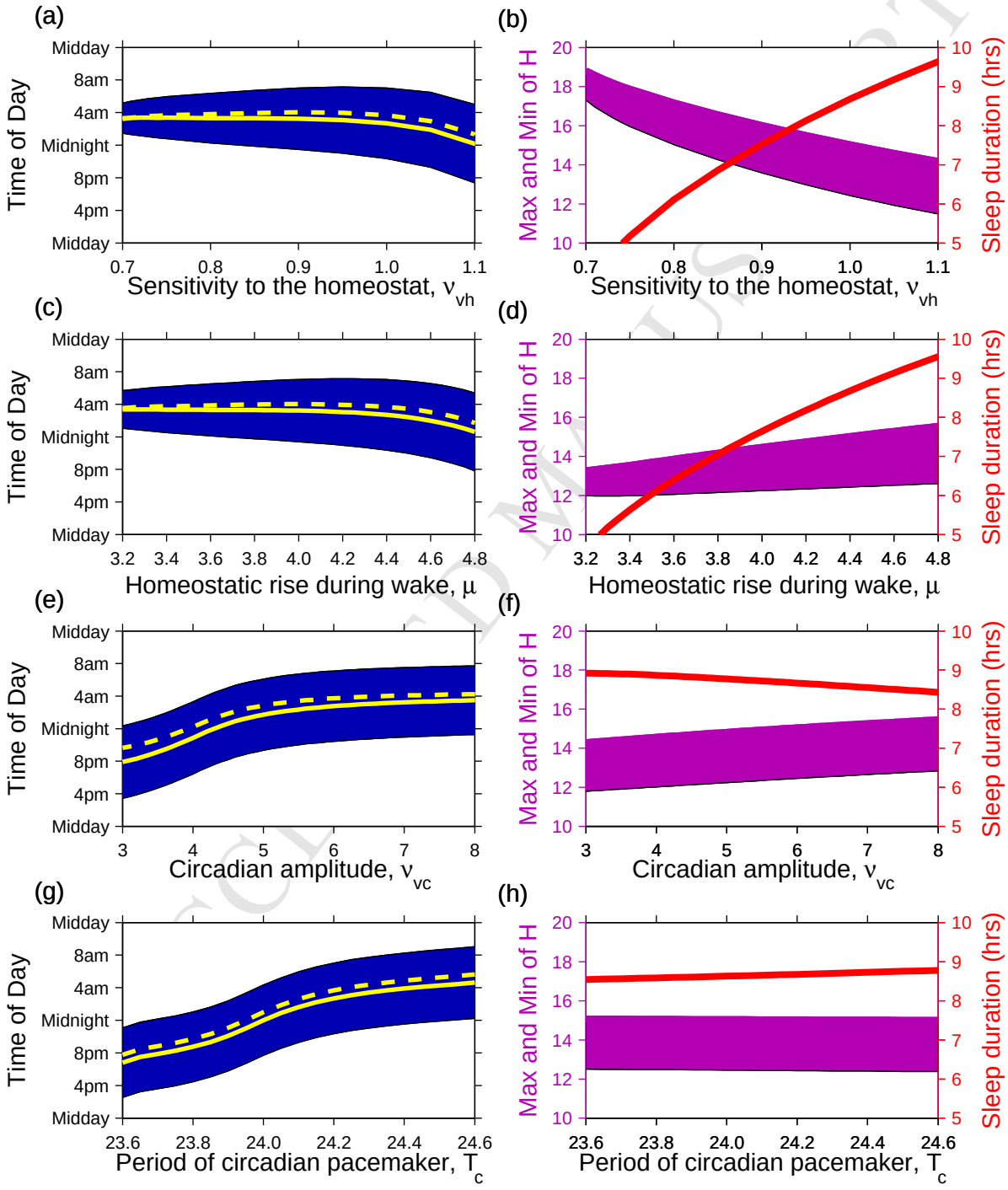

(a)
(b)
(c)
(d)
(e)
(f)
(g)
(h)
Time of Day
Midday
8am
4am
Midnight
8pm
4pm
Midday
Max and Min of H
Sleep duration (hrs)
Sensitivity to the homeostat, ν vh
Homeostatic rise during wake, μ
Circadian amplitude, ν vc
Period of circadian pacemaker, T c

FIGURE 4

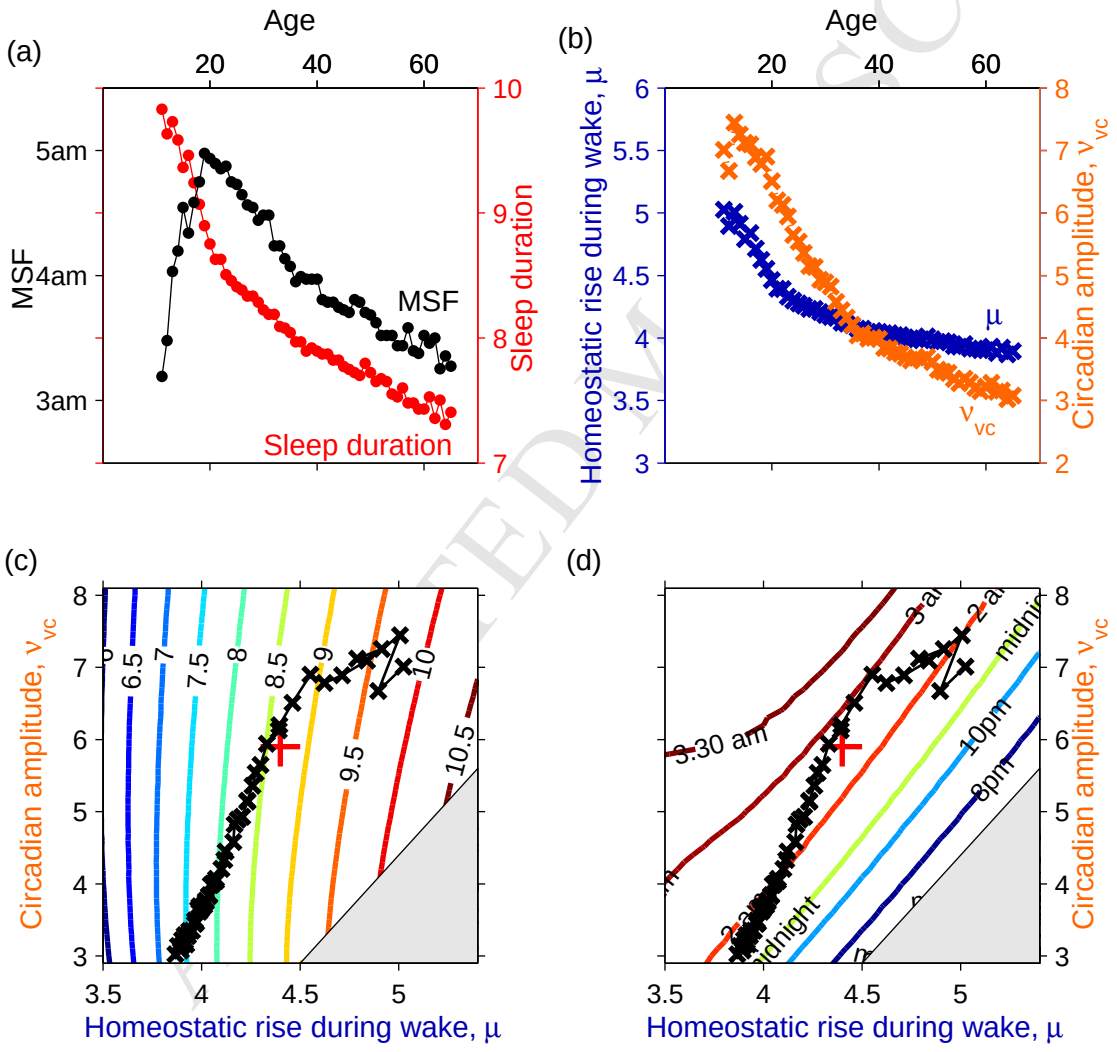

(a)
Age
20
40
60
5am
4am
3am
MSF
Sleep duration
10
9
8
7
MSF
Sleep duration
(b)
Age
20
40
60
Homeostatic rise during wake, μ
6
5.5
5
4.5
4
3.5
3
μ
ν vc
Circadian amplitude, ν vc
8
7
6
5
4
3
2
(c)
Circadian amplitude, ν vc
6.5
7
7.5
8
8.5
9
9.5
10
10.5
3.5
4
4.5
5
Homeostatic rise during wake, μ
(d)
9:30 am
midnight
10pm
8pm
Circadian amplitude, ν vc
3.5
4
4.5
5
Homeostatic rise during wake, μ

FIGURE 5

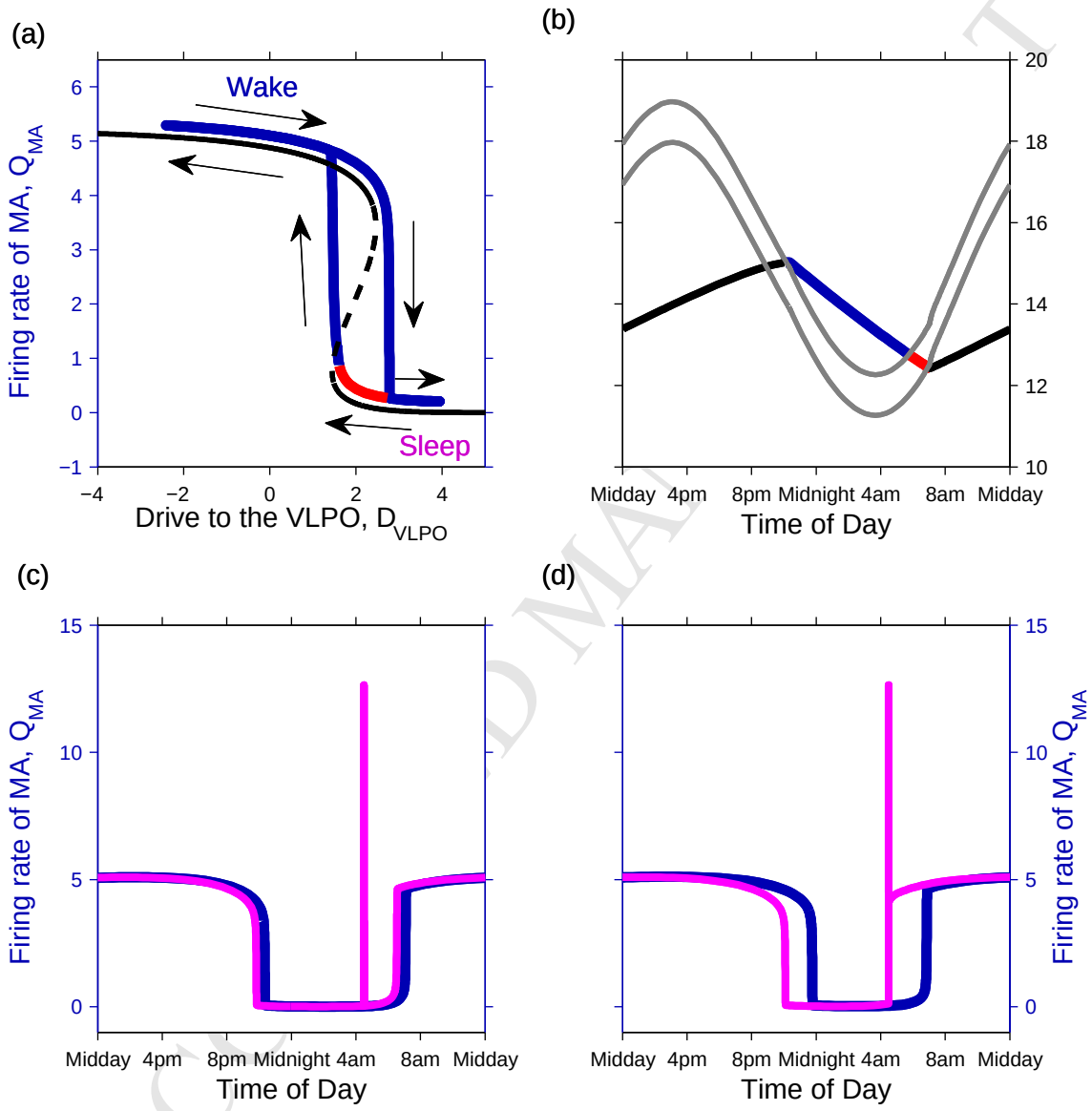
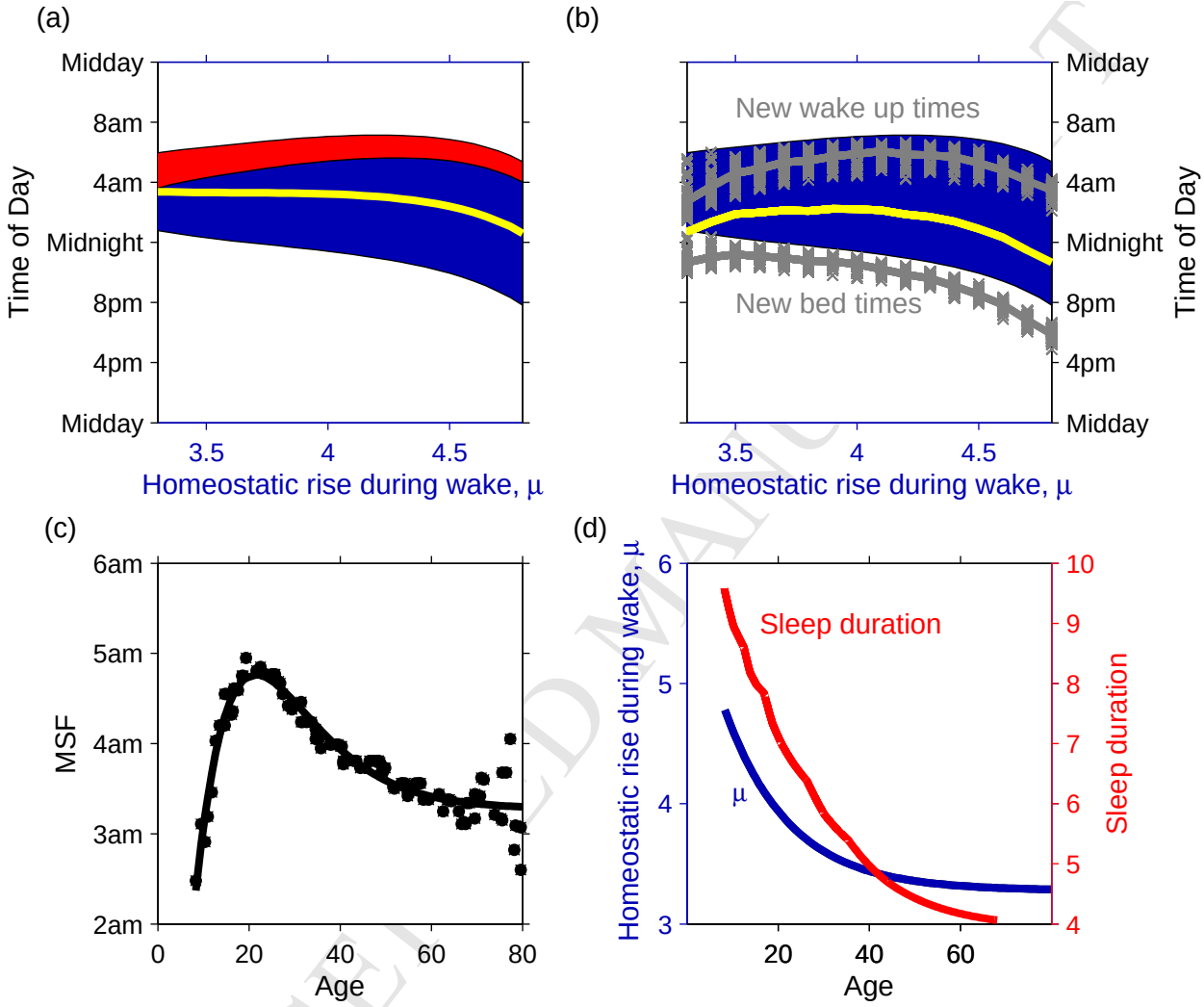

(a)
Wake
Sleep
Firing rate of MA, Q_{MA}
Drive to the VLPO, D_{VLPO}
(b)
Midday 4pm 8pm Midnight 4am 8am Midday
Time of Day
(c)
Firing rate of MA, Q_{MA}
Midday 4pm 8pm Midnight 4am 8am Midday
Time of Day
(d)
Firing rate of MA, Q_{MA}
Midday 4pm 8pm Midnight 4am 8am Midday
Time of Day

FIGURE 6



ACCEPTED MANUSCRIPT